

Dissociative Disorders

Dissociative disorders are complex psychiatric disorders in which individuals experience disruptions in their sense of self, memory, emotions, body, and the sense of continuity with their surroundings. In these disorders, individuals may sometimes feel as though they are observing themselves from outside their own body, may feel that their body or emotions do not belong to them, may perceive their surroundings as unreal, or may have difficulty accessing certain memories. Particularly in conditions associated with traumatic experiences, these symptoms are considered to function as a form of psychological defense that helps the individual mentally distance themselves from overwhelming emotional experiences.

Understanding dissociative disorders requires consideration of the interactions among the brain systems involved in self-processing, emotion, memory, and cognitive control. Functional neuroimaging studies have demonstrated that brain activity in individuals with dissociative disorders differs from that of healthy individuals and that these differences are not confined to a single brain region. In particular, alterations have been reported in regions such as the prefrontal cortex, anterior cingulate cortex, limbic system, amygdala, hippocampus, parahippocampal gyrus, insula, and thalamus.

In depersonalization, one notable finding suggests that the difficulty may lie less in recognizing one's own face than in processing the sense that the face "belongs to oneself." Although an individual may know that the person they see in the mirror is themselves, they may nevertheless experience themselves as unfamiliar, distant, or emotionally disconnected. In fMRI studies examining the processing of one's own face compared with unfamiliar faces, increased activation in the right anterior cingulate cortex and medial prefrontal cortex has been observed, suggesting a possible conflict between conscious and unconscious aspects of self-processing. The finding that the severity of depersonalization is associated with increased activation in the medial prefrontal cortex further supports this relationship.

Another important finding is the reduction in limbic responses to emotional facial expressions. Reduced activation in the amygdala and other limbic structures may be associated with difficulty experiencing emotions in a typical manner or with a sense of emotional distance. Therefore, dissociation can be understood not merely as "feeling different," but also as a disruption in the integration of emotional experience, self-perception, and bodily experience.

In Dissociative Identity Disorder (DID), these differences appear to form a more pronounced and complex pattern. Studies have shown that the brain does not function in exactly the same way across different identity states; rather, different patterns of brain activation may occur

between trauma-related identity states and apparently normal identity states. During the re-experiencing of traumatic memories, alterations have been observed in regions such as the amygdala, insula, parahippocampal gyrus, and caudate. In some trauma-related identity states, decreased activation has also been reported in the prefrontal cortex and regions associated with working memory. Conversely, the observation of increased activation in certain prefrontal and parietal regions during neutral states suggests that the dissociative brain is not simply "less active," but rather functions differently depending on the mental state and task involved.

These findings are particularly important for understanding the relationship between trauma, memory, and emotional regulation. For example, the hippocampus and parahippocampal regions are involved in autobiographical memory, while the amygdala plays a role in processing emotional and threat-related information, and the insula is involved in the awareness of internal bodily sensations and emotional states. Alterations in the activation of these systems may help explain why traumatic experiences can sometimes be intensely re-experienced, while at other times they may feel disconnected, distant, or inaccessible to the individual.

In conclusion, it would be inaccurate to oversimplify dissociative disorders as "a

loss of contact with reality” or “forgetting things.” Current neuroimaging findings suggest that the brain networks involved in fundamental subjective experiences such as “Who am I?”, “Does this body belong to me?”, “Is this emotion really mine?”, and “Does this memory belong to me?” may function differently in individuals with dissociative disorders. However, because most of the existing studies involve small sample sizes and methodological differences, these findings should not be interpreted as establishing a definitive cause-and-effect relationship. Nevertheless, the overall picture is highly significant: Dissociation is not merely a mental state of “forgetting”; rather, it is a complex neurobiological condition associated with alterations in the processes that allow an individual to experience their sense of self, emotions, body, and memories as an integrated whole.

References:

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